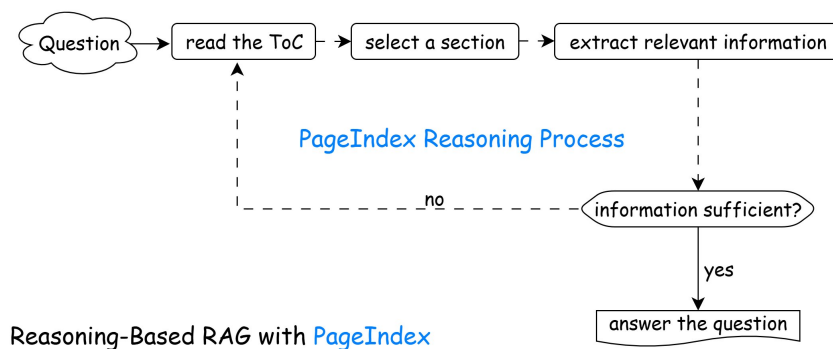


Motivation

- Query and knowledge space mismatch
 - Vector retrieval assumes that the most semantically similar text to the query is also the most relevant. Which is not always true.
- Hard chunking breaks semantic and contextual integrity.
- Hard to deal with in-document reference
 - Like “see Appendix G” or “refer to Table 5.3”.
 - Since these references don’t share semantic similarity with the referenced content, traditional RAG misses them unless additional preprocessing.

General Flow

1. Read the Table of Contents (ToC)
2. Select a Section
3. Extract Relevant Information
4. Is the Information Sufficient?
 - Yes → Process to Answer the Question
 - No → Return to Step 1 and repeat the loop with another section.
5. Answer the Question



1. "Table of Contents (ToC)" Index for LLMs

- Use **JSON-based hierarchical structure** to represent **ToC**
- **ToC** acts as an **index tree**, organizes content into hierarchical
- **Nodes** represent a logical section (e.g. chapter, paragraph, page)

```
Node {  
  node_id: string,           // Unique node identifier  
  title: string,             // Human-readable label or title  
  description: string,       // Optional detailed explanation of the node  
  metadata: object,          // Arbitrary key-value pairs for context or attributes  
  sub_nodes: [Node]         // Array of child nodes (recursive structure)  
}
```

- **node_id**: Serves as a reference key to locate the corresponding raw data.
- **sub_nodes**: Field allows recursive nesting, forming a full ToC tree.
- **metadata**: Field can store semantic information (e.g. document type, author, timestamp, or relevance scores.)

• ToC Example:

```
...  
{  
  "node_id": "0006",  
  "title": "Financial Stability",  
  "start_index": 21,  
  "end_index": 22,  
  "summary": "The Federal Reserve ...",  
  "sub_nodes": [  
    {  
      "node_id": "0007",  
      "title": "Monitoring Financial Vulnerabilities",  
      "start_index": 22,  
      "end_index": 28,  
      "summary": "The Federal Reserve's monitoring ..."  
    },  
    {  
      "node_id": "0008",  
      "title": "Domestic and International Cooperation and Coordination",  
      "start_index": 28,  
      "end_index": 31,  
      "summary": "In 2023, the Federal Reserve collaborated ..."  
    }  
  ]  
}  
...  
}
```

- **start_index / end_index**: Start page / End page
- **summary**: Summary of indicated pages
- Each node in ToC links to corresponding **raw content** (e.g. text, images, tables)
`node_id` → `node_content` (raw content, extracted text, images, etc.)

- This mapping enables the LLM to **select and retrieve** specific nodes as needed.
- **JSON-based ToC index** resides within the LLM's active reasoning context.